II SEM PHYSICS

Astronomy

Time: 2 hrs./week + 01 Hr tutorial Max Marks:
Content Hrs
Unit – 1 -History and Introduction
Chapter 1
Ancient Astronomy
Greek Observations, Sumerian Observations,Mayan Observations, Arabic Observations ,Chinese Observations
2
Chapter 2
Indian Astronomy
Vedic Astronomy, Ancient Astronomy – Aryabhata, Varahamihira, Bhaskara Astronomy in Indian Scriptures, Precession of the Equinox,Celebrations of Equinox
2
Chapter 3
Medieval & Modern Astronomy
Invention of Telescopes, Models of the Solar System & Universe, Observations by Tycho Brahe, Kepler, Galileo, Herschel and Other,Modern Astronomy
2
Chapter 4
Optical tools for Astronomy
Pin Hole, Binoculars, Telescopes & Imaging.
1
Chapter 5
Mathematical Methods of Observations
Angular Measurement, Trigonometric functions, Stellar Parallax
1
Chapter 6 Observational Terminologies
Cardinal Directions, Azimuth, Altitude, Measurements using Compass and Hand. Equatorial Co-ordinates, Light years, Magnitude, Colors etc.
2
Unit – 2: Unit 2: Observations of the Solar System
Chapter 7.
The Sun
Ecliptic and the Orientation of the Earth, Seasons - Solstices and Equinox, Observations of the Sun from Earth during seasons. Eclipses, Zero-shadow day, Sunspots
1
Chapter 8
The Moon
Earth-Moon system – Phases, Lunar Eclipses, Ecliptic and Lunar Orbital Plane – Nodes, Lunar Month, Full Moon Names
1
Chapter 9.
Inner Planets: Mercury & Venus
Observational History, Observational Windows, Appearance, Apparitions,

Elongations, Superior Conjunctions, Inferior Conjunctions, Transits.
2
Chapter 10
Outer Planets
Outer Planets: Mars, Jupiter & Saturn
Observational History.Observational Windows, Appearance, Frequency of Oppositions Oppositions, Conjunctions, Moons Eclipses.Galilean Moons, Saturn's Rings
2

Unit III Major Astronomy Observations
Chapter 11
March to June
Prominent Stars and Constellations Visible during this period, Methods of Spotting.
2
Chapter 12
June to September
Prominent Stars and Constellations Visible during this period, Methods of Spotting.
2
Chapter 13
September to December
Prominent Stars and Constellations Visible during this period, Methods of Spotting.
2
Chapter 14
December to March
Prominent Stars and Constellations Visible during this period, Methods of Spotting.
2

Reference Books:

1.The Stargazer's Guide - How to Read Our Night Sky by Emily Winterburn
2. A guide to the Night Sky – Beginner's handbook by P.N. Shankar
3. The Complete Idiot's guide to Astronomy by Christopher De Pree and Alan Axelrod

Text Books

1. P. N. SHANKAR A GUIDE TO THE NIGHT SKY
https://www.arvindguptatoys.com/arvindgupta/nightskyshankar.pdf
2. BimanBasu , Joy of Star Watching , National Book Trust of India 2013

References Books

Christopher De Pree :The Complete Idiot's Guide to Astronomy, Penguin USA, 2008
Emily Winterburn ,The Stargazer's Guide: How to Read Our Night Sky, Constable and Robinson, 2008

Activities

Sl No Experiment
1 Measuring Seasons using Sun's Position.
2 Measuring Distance using Parallax
3 Estimation of the Stellar Diameter using Pin Hole
4 Measuring Height of an Object Using Clinometer.
5 Star spotting using constellation maps

6 Constellation spotting using Skymaps
7 1. Estimation of ‘Suitable Periods’ to observe deep sky objects using Planisphere.
8 Estimation of the Size of the Solar System in using Light Years.
9 Identification of Lunar Phases across a year.
10 Measuring Constellation of the Sun using Night Skymaps or Planispheres.onomy